Enzymes: Nature's Catalysts

How proteins speed up the chemistry of life

1. What is an enzyme?

Enzymes are biological catalysts: they make chemical reactions happen faster without being used up. Almost all enzymes are proteins (a few are RNA). Without enzymes, reactions in our cells would be too slow to keep us alive — a meal would take years to digest.

TWO IMPORTANT FACTS

- An enzyme makes a reaction **faster**. It does **not** change how much product is made at the end: the final balance (equilibrium) of reactants and products stays the same.
- An enzyme is **specific**: each enzyme works on only one substrate (or a few similar ones).

How do enzymes speed up reactions?

Every reaction needs a push to get started — an energy "hill" called the activation energy, which is linked to a high-energy transition state. Enzymes lower this hill, so many more molecules can get over it every second.

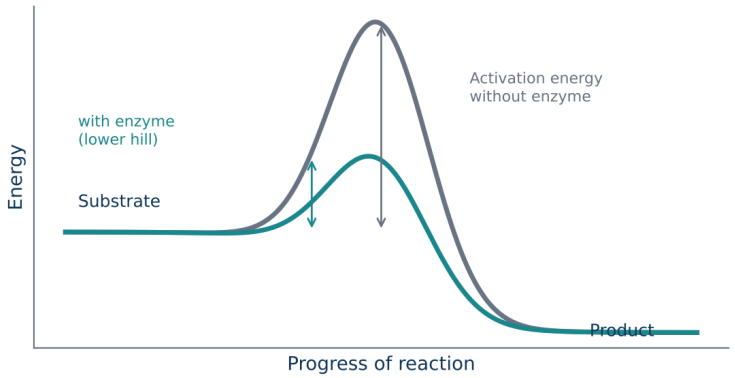


Figure 1. An enzyme lowers the activation energy; the energy of the substrate and the product stays the same.
Source: Own illustration

The active site

Each enzyme has a small pocket called the active site. The molecule it works on, the substrate, fits into the pocket to form an enzyme–substrate complex. The reaction happens, the product is released and the enzyme is free to work again.

- **Lock-and-key model:** the active site has exactly the shape of the substrate — like a key in a lock.
- **Induced-fit model:** the enzyme changes shape slightly when the substrate binds, to grip it better — like a glove closing around a hand. This is closer to reality.

2. What affects enzyme activity?

Because enzymes are proteins, anything that changes their shape changes how well they work.

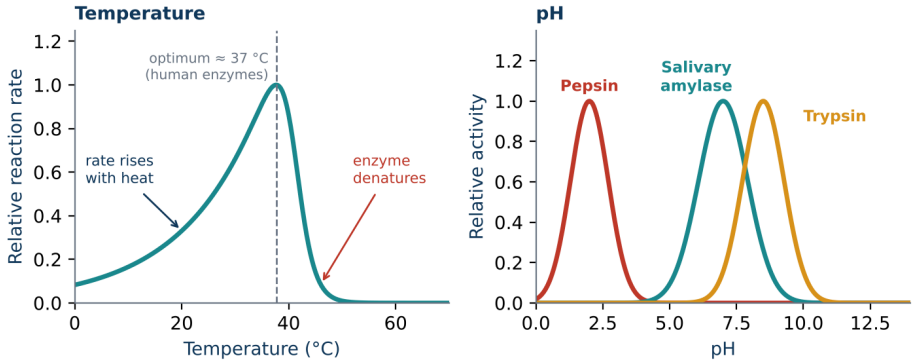


Figure 2. Left: rate rises with temperature until the enzyme begins to denature. Right: each enzyme has its own best pH. *Source: Own illustration*

Factor	Effect
Temperature	Rate rises with heat (more collisions) up to an optimum (about 37 °C for human enzymes). Above that, the protein unfolds (denatures) and the enzyme stops working, usually for good.
pH	Each enzyme has an optimum pH. Away from it, charges on the protein change and the active site loses its shape.
Substrate concentration	More substrate gives a faster rate, until all the enzyme molecules are busy (saturation).
Enzyme concentration	More enzyme gives a faster rate, as long as there is enough substrate.
Inhibitors	Molecules that slow or stop an enzyme. *Competitive* inhibitors look like the substrate and block the active site; *non-competitive* inhibitors bind elsewhere and change the enzyme's shape. Many medicines (and poisons) are inhibitors.
Activators / regulation	Cells switch enzymes on and off with activators and inhibitors, so they make only what they need.

Helpers: cofactors and coenzymes

Many enzymes need a non-protein helper to work. Cofactors are metal ions (such as zinc or iron) or small organic molecules. Organic helpers that bind loosely are called coenzymes; those bound tightly are prosthetic groups. Many coenzymes are made from vitamins in our diet — which is one reason we need vitamins.

Isozymes (isoenzymes) are different forms of an enzyme with slightly different amino-acid sequences that catalyse the same reaction.

3. Examples

Enzyme	Where	What it does
Amylase	Saliva, pancreas	Breaks starch into sugars (why bread tastes sweeter if you chew it for a while)
Pepsin	Stomach	Digests proteins; works at very low pH (about 2)
Lactase	Small intestine	Splits milk sugar (lactose)
Catalase	Almost all cells	Breaks down hydrogen peroxide into water and oxygen
DNA polymerase	Cell nucleus	Copies DNA (Day 1, and used in PCR)
β-galactosidase	Bacteria (lac operon)	Splits lactose (Day 2)
Laccase	Fungi, bacteria, plants	Oxidises many phenol-like compounds; studied for cleaning pollutants from water

Case study: pepsin

Pepsin digests proteins in the stomach, which is very acidic, so pepsin works best at a pH of about 2. It is made as an inactive form, pepsinogen, by the stomach's chief cells. Hydrochloric acid, released by parietal cells in response to the hormone gastrin (from G cells), converts pepsinogen to active pepsin. This protects the cells that make it from being digested themselves. When the stomach contents move into the small intestine, bicarbonate from the pancreas neutralises the acid, protecting the intestinal wall — and pepsin stops working at the higher pH.

4. Demonstration: catalase in a potato

SAFETY
Wear goggles. Use only 3 % hydrogen peroxide (household strength) and do not drink it. Wash off splashes with water.

Aim: to show an enzyme at work and to see how heat affects it.

1. Label three clear cups: **A raw**, **B boiled**, **C cold**. Put a few small pieces of raw potato in A, boiled-then-cooled potato in B, and potato kept in the fridge in C.
2. Pour the same amount of hydrogen peroxide (about 2 cm deep) into each cup.
3. Watch for **bubbles** (oxygen). Measure the height of the foam after 2 minutes.

Cup	Treatment	Bubbles / foam height	Explanation
A	Raw potato, room temperature		

Cup	Treatment	Bubbles / foam height	Explanation
B	Boiled potato		
C	Cold potato		

Expected: raw potato foams the most; boiled potato hardly at all (catalase denatured); cold potato foams less (slower reaction). Ask students to explain every result.

Check your understanding

1. Why does an enzyme not change the amount of product at the end?
2. Why does boiling stop an enzyme from working for good?
3. Why does pepsin work in the stomach but not in the intestine?
4. What is the difference between a cofactor and a coenzyme?